Chapter 13 Review

Campbell Biology, AP Edition, 12e

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Summary of Key Concepts

Why is meiosis considered a successful evolutionary strategy to increase the fitness of eukaryotic organisms? **(BIG IDEAS 1 & 3)**

Concept 13.1: Offspring acquire genes from parents by inheriting chromosomes

- Each **gene** in an organism's DNA exists at a specific **locus** on a certain chromosome.
- In **asexual reproduction**, a single parent produces genetically identical offspring by mitosis. ****Sexual reproduction** combines genes from two parents, leading to genetically diverse offspring.

Explain why human offspring resemble their parents but are not identical to them.

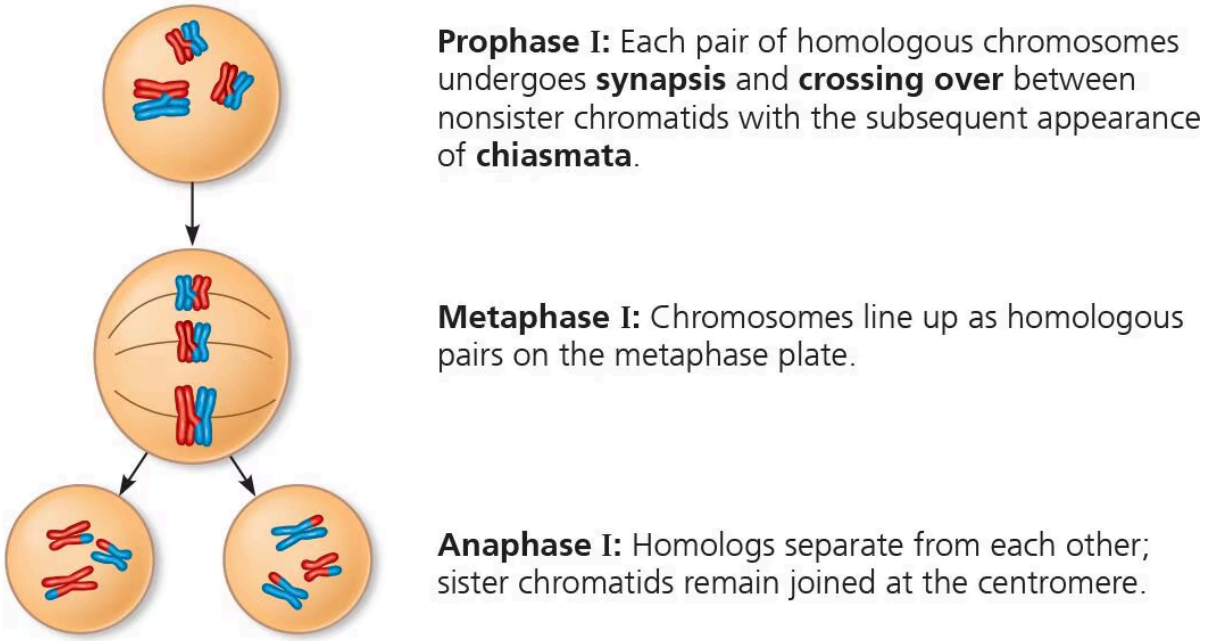
Concept 13.2: Fertilization and meiosis alternate in sexual life cycles

- Normal human **somatic cells** are **diploid**. They have 46 chromosomes made up of two sets of 23 chromosomes, one set from each parent. Human diploid cells have 22 pairs of ****homologs** that are **autosomes**, and one pair of **sex chromosomes**; the latter typically determines whether the person is female (XX) or male (XY).
- In humans, ovaries and testes produce **haploid gametes** by **meiosis**, each gamete containing a single set of 23 chromosomes (n=23). During **fertilization**, an egg and sperm unite, forming a diploid (2n=46) single-celled **zygote**, which develops into a multicellular organism by mitosis.
- Sexual life cycles differ in the timing of meiosis relative to fertilization and in the point(s) of the cycle at which a multicellular organism is produced by mitosis.

Compare and contrast the life cycles of animals and plants.

Concept 13.3: Meiosis reduces the number of chromosome sets from diploid to haploid

- The two cell divisions of meiosis, **meiosis I** and **meiosis II**, produce four haploid daughter cells. The number of chromosome sets is reduced from two (diploid) to one (haploid) during meiosis I.
- Meiosis is distinguished from mitosis by three events of meiosis I:



Prophase 1: each homologous pair undergoes synapsis and crossing over between non-sister chromatids with the subsequent appearance of chiasmata.

Metaphase 1: chromosomes line up as homologous pairs on the metaphase plate.

Anaphase 1: homologs separate from each other; sister chromatids remain joined at the centromere.

- Meiosis II separates the sister chromatids.
- Sister chromatid cohesion and crossing over allow chiasmata to hold homologs together until anaphase I. Cohesins are cleaved along the arms at anaphase I, allowing homologs to separate, and at the centromeres in anaphase II, releasing sister chromatids from each other.

In prophase I, homologous chromosomes pair up and undergo synapsis and crossing over. Can this also occur during prophase II? Explain.

Concept 13.4: Genetic variation produced in sexual life cycles contributes to evolution

- Three events in sexual reproduction contribute to genetic variation in a population: independent assortment of chromosomes during meiosis I, crossing over during meiosis I, and random fertilization of egg cells by sperm. During crossing over, DNA of nonsister chromatids in a homologous pair is broken and rejoined.
- Genetic variation is the raw material for evolution by natural selection. Mutations are the original source of this variation; recombination of variant genes generates additional genetic diversity.

Explain how three processes unique to sexual reproduction generate a great deal of genetic variation.